Participant Publication/Subscriber, Readers and Writer Instantiation

The application uses RTI's System Designer with the TMS idl files to standup all entities via XML App Create along with RTI's Dynamic Data feature.

Topics and Implementation Plans

The following is a list of 19 different topics and the way this design intends to address them.

TMS Keyed SampleId and DDS Resource usage - As defined in the TMS specification, SampleId is always a unique combination of DeviceId plus a monotonically incrementing Sequence Number. This means that each sample will take a unique memory resource (receiver cache) vs. placed as a sample in the same cache. This will require a DDS Dispose operation after each request is received and dispatched.

Reader Topic filters

Reader topics are keyed by SampleID, which is comprised of the DeviceID and a "SequenceNumber." The DeviceID is unique to each device and retrieved at runtime via your devices internal interface. It will be used to programmatically apply filters to the 5 reader topics defined below so that the device needs only concern itself with requests and responses that are directed to *this* device and not deal with all such traffic in the system.

With the current list of supported Topics, RequestResponse will have 4 Topics (1 Subscriber and 3 Publishers associated with the 1 Command Publish and 3 Command Subscriber topics listed in Appendix B.)

General Topic Design - There are five design patterns associated with the 19 identified topics resulting in 13 required threads. They are identified as follows:

Published Topics (14) and Patterns (5) (and threads (8 w/ 3 optional))**: (x, y, z notation shown)**

> **Publish Once (3, 1, 3 (optional))** - A topic that is published once prior to the main loop with Publish Last QoS Profile enabling late joiners. These include `DeviceAnnouncement, DevicePowerPortList, and Device Grounding`. Because these are published only once in the main loop, attaching a Writer Event monitoring thread is optional.
>
> **Periodic Publish(3, 1, 3)** - A topic, once enabled, is sent out at a regular interval. These include the three defined as Fixed Rate (`Heartbeat, DevicePowerMeasurementList, EngineStart.`
>
> **On Change Publish (4, 1, 4)** - A topic that is its status changes is published. These include `DevicePowerStatusList, LoadSharingStatus, SourceTransitionState, and FingerPrintNickname` topics.
>
> **In Response Publish(3, 1, 0)** - Request Response to a received Command Topic. These are published within the associated subscriber's thread context.
>
> **Command Request Publish(1, 1, 1)** - In this example, the `MicrogridMembershipRequest` will be sent at startup. Presumably, it would

normally be sent based on some operator control (i.e., powering up the device). Note this topic is actually defined as an Advisory topic, meaning the generator may join the grid without permission (1st paragraph page 208 of Draft TACTICAL MICROGRID 12 COMMUNICATIONS AND CONTROL document dated 1 February 2021.) While advisory, to be a good citizen, this design will respect the `MicrogridMembershipApproval` Response `MMR_COMPLETE`. `MMR_COMPLETE` will be used gate (enable and disable) periodic and On Change Publish topics.

Receive Topics (5) and Patterns[1] (1) (and 5 threads)**:**

All 5 Receive Topics fit within a single receive pattern. These include the three Commands Expected to be sent to this device: `LoadSharingRequest`, `SourceTransitionRequest`, and `FingerprintNicknameRequest`. Each of these requires a simple `RequestResponse` to be sent with the `DeviceID` and `SequenceNumber` of the request populated in the response along with the response code. The fourth received `MicrogridMembershipOutcome` sent as a result of an on-change from the controller. The fifth received topic is the `RequestResponse` associated with the `MicrogridMembershipRequest`. The sequenceNumber within the SampleID is used to match the response to a given request. This `RequestResponse` may be used to update a status panel on the device for convenience, but no action results or are necessary since in the current design `MicrogridMembershipRequest` continuing to be periodically sent until `MicroGridMembershipApproval` is `MMR_COMPLETE`.

TMS / DDS interaction with your internal system variables:

Generally, a TMS request, or local operation, results in an internal system variable change. When a TMS request, the variable change should be done in the TMS request command thread context. An equivalent TMS “shadow variable” should also be maintained, but not changed at the time of the internal system variable modification. The main loop should be continually updating non-“OnChange” TMS variables and checking for internal “OnChange” variables to be different than the TMS shadow variables. If this is detected, the variables should be made equivalent and an OnChange TMS message should be triggered.

Thread Interaction

The only thread interaction involves the five receive topics as follows:

- The three Command Requests - can indirectly cause an On Change event with the triggers an OnChange publish topic.
- `MicroGridMembershipApproval MMR_COMPLETE` (Or not) will gate the enable/disable functions associated with all periodic or On change topics.

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[1] This could be considered two design patterns, a straight Receiver and a Receiver that has a common code block to generate the response (In implementation, likely just pass in a ‘send response flag’ in the thread control block.)

- `MicrogridMembershipRequest` RequestResponse may optionally update a localized display panel on the device.

Lock / Atomic Operation Considerations

The current threading model is simple and has little interaction between threads. The design has no thread-safety / locks or synchronization between threads. There may be On Change Data that must be consistent among data members associated with a given topic before being presented to the controller (i.e., published). Care should be taken to ensure all of the On Change topic data is consistent before triggering the associated On Change topic publication. If this is an issue, data should be updated via multiple non-atomic internal operations within the main loop. After specific internal updates and variables are changed, any TMS related variables that require a change should trigger On Change topic publication. If needed, all of the variables should be copied out to the topic, and the topic On Change triggered.

Heartbeat

The current heartbeat runs independently as a periodic thread. If this is all the Tactical Microgrid Spec wanted, they could have simply required DDS liveliness to be QoS enabled and not bothered with the Heartbeat topic. For a Heartbeat to be meaningful, all the other threads, including the main thread in the application should checkin (set a semaphore). The Heartbeat then checks to ensure that all other threads have checked in with it. Upon failure the Heartbeat topic would disable itself.

To implement a meaningful Heartbeat generically (less tms specificality, all threadInfoBlocks constructors take a `hbDeadmanPeriod` defaulted to `DDS_DURATION_INFINIT` (Thread Monitoring off). This value is used as the waitset duration. When ever a waitset triggers due to a time out, a global heartbeat semaphore array indexed by topicEnum is set. By using a `hbDeadmanPeriod` half as long as a heartbeat period, all topics in the the semaphore array should be set for each heartbeat period. The heartbeat handler then checks all monitored threads for 3 missing heartbeats. If any monitored thread has timed-out the heartbeat thread disables itself and stops sending heartbeats.

Response and Request Sequence handling

As defined, requests and responses are required to have a keyed SampleId composed of the deviceId (Fingerprint) and a sequence number that monotonically increments with any request. The device unique Sequence number allows this device to correlate responses to requests (especially if there is more than one outstanding - which is always possible if a request or response is lost.

The request and response pattern is both sent and received. As a generator, we will both issue requests and receive them and need to ensure correlation with the response received or sent respectively.

Sent Responses: Correlating sending responses to specific requests is fairly easily handled. Here, each response will be issued in the context of the request. That is, upon receiving a request, the handler thread for that request will turn the response around within the handler thread as it receives the request (this handler may also do work within the device to carry out the request - which in turn may change some data that will trigger an On-Change topic to be issued.

Received Responses. Responses are generally used to acknowledge the receipt of a command and might be used to post the state of the response to an operator display. Response correlation cannot be done in the context of the sending request thread (since we have no way to know if the request got lost and may want to issue a new request). This design will keep the last ten request sequenceNumbers, with the enum of the topic (a paired ReqQEntry) in a circular queue to handle correlation. When a response arrives, it will be modulo looked upon the circular queue. If it's not there, the response will be dropped and ignored. If it is in the circular queue, it can be correlated with the command and optionally used to update a display (or potentially take other programmatic operations like retry etc.)

Further this enqueueing of the requested sequence number and topic_enum will be done by the `RequestSequenceNumber::getNextSequnceNo` (enum TOPICS_E) function (used when writing the request topic sample in the main loop). To ensure only this function performs the enqueueing, the `ReqCmdQ` write function is private and the `RequestSequenceNumber` class has been declared a ‘friend’ object (giving it private access). This means that a sequence being written to the `ReqCmdQ` can not be a response being processed (read from the queue) since the `Request topic` sample for that sequence has not been written yet. By writing the `sequence_number` to the queue entry before the enum and reading the `sequence_number` last, if the sequence number matches the request_response we are processing we necessarily have a paired enum. The only race condition that could occur is that we get an enum that would have paired, but is overwritten by the main_loop request topic being sent, returning us a sequence that does not match our `request_response sequence.` In this case we discard the response as designed (for further information read all the comments around these two objects, `RequestSequenceNumber` and `ReqCmdQ`.)

Note: In the lower level ReqCmdQ entry structure I added a `responseNotProcessed` flag, initialized to False (meaning the entry has been processed). The `RequestSequenceNumber::getNextSequnceNo` will call a `ReqCmdQ::reqCmdQWrite` which will set this flag true. `ReqCmdQ::reqCmdQWrite` will also check this flag, and if true, meaning no response for the sequence number was processed it will print out a message stating that it is overwriting an outstanding requestResponse. The previous sequence number is first locally stored and the requesterEnum has not yet been overwritten. So an application can do more with this information if needed. This flag is cleared upon when ever the ReqCmdQ is read and a match is made (meaning a response for an outstanding request was found and returned to the application - i.e. processed). Because the reader, writes a piece of data (clears the `responseNotProcessed` flag) and there is no thread safety around this code, there may be a very small possibility that IFF a requestResponse is missing (which should not occur with DDS recommended ReliableReliabilty QoS for Commands) AND the response show up exactly RQ_SIZE requests later AND is is perfectly time, the message of Overwriting a processed Response in RequestResponseQ may be incorrect (i.e. the entry was processed and it detects it was not) or not detected when it should have. (I think scenario is just about impossible to occur, even if Best Effort QoS was used).

Appendix A - Relevant Data Structures (Reference)

```
struct ReplyStatus {

    /** Indicator of success or failure. Intended to support automatic
handling. Values from 100 to 599 are as defined in the IETF RFCs.
Values outside that range are reserved for future versions of this
standard. Selected values are defined as constants. */
    unsigned long code;

    /** Short textual description of the status code intended for
human operators. */
    string<MAXLEN_reason> reason;
}; // struct ReplyStatus

/** The request has succeeded. */
const unsigned long REPLY_OK = 200;

/** Some value in the request was invalid. */
const unsigned long REPLY_BAD_REQUEST = 400;

const unsigned long REPLY_METHOD_NOT_ALLOWED = 405;

const unsigned long REPLY_CONFLICT = 409;

const unsigned long REPLY_GONE = 410;

const unsigned long REPLY_PRECONDITION_FAILED = 412;

const unsigned long REPLY_REQUEST_ENTITY_TOO_LARGE = 413;

const unsigned long REPLY_INTERNAL_SERVER_ERROR = 500;

const unsigned long REPLY_NOT_IMPLEMENTED = 501;

const unsigned long REPLY_SERVICE_UNAVAILABLE = 503;

/** Request is valid, and authorization is required before processing.
*/
const unsigned long REPLY_PENDING_AUTHORIZATION = 600;


struct SampleId {

    /** Identity of the device that sent this message. */
    Fingerprint deviceId;

    /** A number that is unique to each SampleId sent by this device.
It commonly starts at 0 and increments for each generated SampleId,
but such behavior is not required. For example, a pseudo-random
sequence may be used to avoid leaking information about how many
identities have been generated. */
```

```
    unsigned long long sequenceNumber;
}; // struct SampleId


struct RequestResponse {

    /** Copy of the corresponding requestId. */
    @key SampleId relatedRequestId; //@key

    /** Indication of success or failure. */
    ReplyStatus status;
}; // struct RequestResponse

enum MicrogridMembership {
    MM_UNINITIALIZED, // Uninitialized value.
    MM_JOIN, // Join or stay in the microgrid.
    MM_LEAVE // Leave or stay out of the microgrid.
}; // enum MicrogridMembership

struct MicrogridMembershipRequest {

    /** Identity of this request */
    @key SampleId requestId; //@key

    /** The device requesting a membership change. May be a
platformId, indicating that all deviceIds on the platform are
affected. */
    Fingerprint deviceId;

    /** The desired membership state. */
    MicrogridMembership membership;
}; // struct MicrogridMembershipRequest

enum MicrogridMembershipResult {
    MMR_UNINITIALIZED, // Uninitialized value.
    MMR_REPLACED, // New request received before preparation for this
request was complete. Device should discard this request and wait for
the new request to complete.
    MMR_COMPLETE, // Completed preparation for the requested action.
No significant issues were encountered.
    MMR_BLOCKED // Preparation could not be completed without a
significant negative impact such as shedding load. Operator should
resolve the issue before proceeding.
}; // enum MicrogridMembershipResult

struct MicrogridMembershipApproval {

    /** Identity of this request */
    @key SampleId requestId; //@key

    /** Copy of the corresponding
MicrogridMembershipRequest.requestId. */
    SampleId relatedRequestId;
```

```
    /** Device (or platform) that has been approved. */
    Fingerprint deviceId;

    /** Requested state. */
    MicrogridMembership membership;

    /** Indicate whether the microgrid is ready to complete this
request. */
    MicrogridMembershipResult result;

    /** Short, human-readable text description of a blocked request.
This should summarize to the operator what the issue is (e.g.
``blocked by interruptible load'', ``blocked by critical load'', or
``manual action required''). */
    string<MAXLEN_hint> hint;
}; // struct MicrogridMembershipApproval
```

Appendix B - List of Example Topics
(With Filter Param for Device)

For subscribed / reader topics on the device, we only want topics directed to us (our deviceId). Since we only know our deviceId at runtime, we create the filter expression and default params using System Designer and then programmatically change the filter parameters as we find the base readers. For the SIM MSM we might only filter reqResponses that are directed to us.

1. ~~DeviceAnnouncement (once, durability)~~ - Device filter N/A, MSM filter None/NULL
   - Device Pubs Only - would be ok to filter form me. Filter deviceId == me.
   - `this_device_id in field "deviceId"`
   - In example SIM does not listen but real SIM would want to listen to all announcements

2. ~~Heartbeat (periodic)~~ - Device filter N/A, MSM filter None/NULL
   - Device Pub Only
   - `this_device_id in field "deviceId"`
   - Topic contains no target deviceId, only the source. If a Device needs to listen - it would have to filter on deviceID != Me and could be overwhelmed by many device heartbeats.
   - Sim MSM or device does not listen, MSM would want to listen to all device Heartbeats.
   - Example MSM SIM does not generate heartbeat

3. ~~MicrogridMembershipRequest~~ - Device filter N/A, MSM filter None/NULL
   - Device Pub Only - Elicits RequestResponse from MSM
   - `this_device_id in field "deviceId"`
   - Topic contains no target deviceId, only the source
   - No filter on MSM since MSM is must receive all MembershipRequests.

4. MicrogridMembershipOutcome (from MSM to device) - Device filter - Yes, MSM filter N/A
   - sent "On Change" pattern from MSM - relatedRequestId.deviceId is the requesters ID
   - Outcome/Response/Reply and Result topics all types take the SampleID from associated request type (e.g. fooRequest ->fooOutcome) and copies it to the relatedRequestId.
   - Only want Outcomes / Responses/Replies/Results directed to me
   - ON DEVICE (but any participant that receives this topic): Filter relatedRequestId.deviceId == me.

5. SourceTransitionRequest (sent to device from MSM) -Device filter - Yes, MSM filter N/A
   - Elicits RequestResponse from Device
   - All requests* have the source SampleId requestId and the target Fingerprint deviceId
   - MSM would issue this topic / write and not receive.
   - Only want Requests directed to me
   - ON DEVICE (but any participant that receives this topic): Filter deviceId == me.

   * ChangeNicknameRequest of course uses Fingerprint id;
   * OperatorIntentRequest uses SampleId requestId.deviceId

6. ~~SourceTransitionState~~ - Device filter N/A, MSM filter None/NULL
   - Device Pub Only Sent "On Change" pattern from Device
   - `this_device_id in field "deviceId"`
   - State Topics, in general, have no destination deviceId - just source (any filter !=Me)
   - SM would not want to filter -

7. RequestResponse (Reader)- Device filter - Yes, MSM filter Yes
   - Device Pub

- DeviceId not copied, rather the response id is read from request and copied to the relatedRequestId.deviceId
- Receive from MSM or other devices responding to MSM
- ON DEVICE (but any participant that receives this topic): relatedRequestId.deviceId == me.

8. ~~RequestResponse (Writer).~~ (See receive, sent topics will be received by all parties including the sender without a filter)

Appendix C - "To Do" list

The following is a list of enhancements and optimizations in no particular order.

1. Dispose or delete the key'd requests and responses as they arrive. By TMS definition, they are all uniquely key'd and will continually be added to the reader caches and saved (effectively creating a resource leak).
    - Use Unregister and Dispose API calls

2. Can creating threads/info blocks also be done in a loop - maybe move to OO design (see below)

3. Investigate Objectifying Topics: Thread processing and handlers seems a bit too much c/procedural vs. Threading Objects with generic handlers and inheritance to specific handlers.

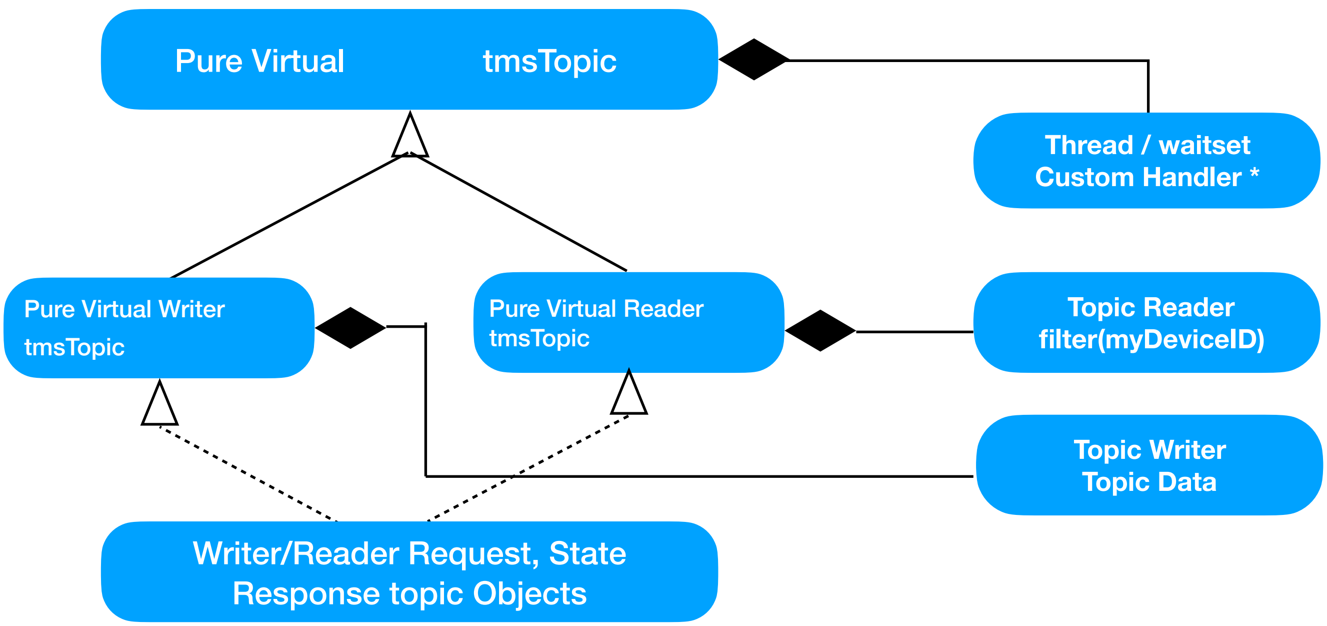
4. ~~Pre-populate all writer topics with this_device_id in the writer handle lookup loop~~.
    - Done for Device Only not Sim MSM (perhaps could repopulate "requestId.deviceId" if we were making a real MSM.

Appendix D - Objectifying

Despite being written in C++, the current code is very C like, using structures to organize the topics vs objects. This may make the process of defining specific topics more rote than an Object Oriented design.

This section will explore how we can Objectify the code. Allowing users to instantiate “request::writer / reader objects or state::writer/ reader objects. Each object would have a thread and a custom, statically registered handler. Writers would each have associated Topic Data and Readers would have Content Filters.

The concrete classes would be instantiated from the class hierarchy shown below and would implement the TMS communications pattern.



Appendix E - Tactical Microgrid Spec Challenges

While implementing the Tactical Microgrid spec, we noted some concerns or areas that could be better defined.

1. Key'd Unique SampleId cause a virtual resource leak unless continually disposed.

2. The spec neither includes or excludes the possibility of multiple outstanding requests by a device to the MSM or MSM to a given device. [As such this application implantation, tracks requests and sequence numbers, matching them up upon receiving a RequestResponse.]

3. Spec should detail 'heartbeat' - Heartbeat could either be done with DDS liveliness or should require any threads (including the main thread) to update a heartbeat mask count. If missed more than 3 times the heartbeat should disable itself. As it sits now I can spawn a 'heartbeat' thread that mindlessly sends heartbeats regardless of the applications ability to service the system.

4. DeviceId management - DeviceIds need to be unique within in a Microgrid system but not universally unique (i.e. two devices with the same DeviceID could be each in a different microgrid.) As currently implemented, it appears that to ensure they are globally unique and manufacturable, some DeviceID authority will have to dole out Ids so device manufactures can populate the deviceId in the manufacturing EPROM.

   An Alternate approach - use the DDS GUID as the deviceId or even better a NIC Mac address as the DeviceID or a base for the deviceId. They are guaranteed to be unique within a microgrid instance and can be fetched at runtime and/or filtered upon. Using a MAC is preferred since it is globally unique and could be used for record/replay on another simulated network without worry of collision and or subject to format change (both of these possibilities is extremely unlikely but a MAC would be more stable.)